Umar Al Mukhtar Ibrahimkutty

Full Stack Developer | Backend Engineer | AI Systems Developer
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PROFILE SUMMARY

Computer Science undergraduate with experience building production-scale backend systems, AI-integrated applications, and full-stack web platforms. Developed software deployed for Smart City operations, engineered backend infrastructure serving thousands of users, and built an award-winning desktop application for automated video editing. Strong foundation in web technologies, distributed systems concepts, APIs, databases, and software engineering practices.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Java, C

Frontend: React.js, Next.js, Tailwind CSS, Three.js

Backend: Node.js, Express.js, REST APIs, JWT Authentication

Tools & Platforms: Git, GitHub, Vercel, FFmpeg, Tauri, Google Apps Script

Databases: MongoDB, PostgreSQL, MySQL, Prisma, Supabase

ML / CV: YOLOv11, OpenCV, Groq Vision API, LLM Integration

EXPERIENCE

Backend Developer Smart City Project – TVM ICCC *Dec 2024 – Present*

- Developed scalable REST APIs using Node.js, Express.js, and MongoDB for a Smart City waste detection platform deployed at the Thiruvananthapuram Integrated Command and Control Centre.
- Integrated YOLOv11 and OpenCV computer vision pipelines with backend services through asynchronous inference workflows.
- Designed structured logging, monitoring, and error-handling systems to improve production reliability.
- Collaborated with frontend teams to deliver real-time analytics dashboards for city administrators.

Backend Intern Mulearn Foundation *Jan 2026 - Jun 2026*

- Developed and maintained backend API endpoints supporting core platform infrastructure.
- Refactored existing services and contributed to production deployments with the engineering team.
- Participated in code reviews, debugging, and backend architecture discussions.

Web Developer Kasaroam Tourism Platform *Jan 2025 – Present*

- Built component-based frontend features for a Smart City–backed tourism platform, optimising responsiveness and page load performance for production deployment.

PROJECTS

FocusFlow — Top 5 Winner, Codex Community Hackathon 2026 Rust • Tauri • React • FFmpeg

- Built a desktop application that transforms raw screen recordings into professional demo videos through automated click-based zooming and camera movements.
- Engineered recording pipelines using Rust and FFmpeg with automatic timeline generation and post-processing workflows.
- Implemented click tracking, automated editing logic, and export systems for streamlined video production.
- Selected among the Top 5 winning projects in the Codex Community Hackathon 2026.

MatchUndo Next.js • PostgreSQL • Prisma • Supabase

- Designed and launched a community platform connecting football fans for shared match-viewing events.
- Implemented authentication, moderation workflows, analytics tracking, SEO optimization, and venue discovery features.
- Built and deployed the complete product independently from architecture through production release.

Smart City Waste Detection Dashboard Node.js • Express.js • MongoDB • YOLOv11 • OpenCV

- Engineered backend architecture and AI integration for a city-scale litter detection system deployed in production.
- Designed APIs, inference pipelines, and data-processing services supporting real-time monitoring.

MediScribe – Handwritten Prescription Digitiser Python • Flask • Groq Vision API

- Built an AI-powered prescription digitization platform converting handwritten prescriptions into structured medical data.
- Developed image-processing, prompt-engineering, validation, and confidence-scoring pipelines

Inventory Manager Next.js • PostgreSQL • Prisma

- Developed inventory management software supporting purchases, consumption tracking, and asset management.
- Designed full-stack workflows currently used in production by a local business.

LEADERSHIP & ACTIVITIES

Chief Technology Officer — IEDC LBS College of Engineering *Mar 2024 – Dec 2025*

Technology & Platform Head — IEDC Summit 2025 *Sep 2025 – Dec 2025*

Chair — IEEE Sensors Council SBC LBSCEK *Nov 2025 – Present*

Webmaster — IEEE IA/IE/PELS Joint Chapter Kerala Section *Feb 2026 – Present*

EDUCATION

B.Tech – Computer Science and Engineering

LBS College of Engineering, Kasaragod, Kerala
Expected Graduation: *June 2027*

CERTIFICATIONS

- Frontend Developer (React) — HackerRank
- Python Basics — HackerRank
- JavaScript Algorithms and Data Structures — freeCodeCamp
- Front-End Development Libraries — freeCodeCamp
- Responsive Web Design — freeCodeCamp
- Prompt Engineering Certification

ACHIEVEMENTS

- Top 5 Winner — Codex Community Hackathon 2026
- Led technology infrastructure supporting 5,000+ event participants.
- Built and deployed production systems used by Smart City and organizational stakeholders.

INTERESTS

Open Source • Distributed Systems • AI Applications • Software Architecture • Competitive Programming